

A Neural Network-Based DPD Algorithm For Low SNR And Off-Grid

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Abstract—Addressing the degradation of traditional direct position determination (DPD) under low signal-to-noise ratio (SNR) and limited data, this paper proposes a neural network-based direct localization algorithm. The algorithm employs multi-scale features extracted from covariance matrices, enhanced by spatial feature fusion, to directly predict source locations. Unlike multi-label classification, it models localization as a regression task, achieving off-grid, high-resolution estimation. Simulation results demonstrate its superior robustness and accuracy compared to classical multiple signal classification (MUSIC) and conventional convolutional neural networks (CNNs), particularly in low SNR environments.

Index Terms—direct position determination, off-grid, deep learning, regression task.

I. Introduction

Source localization is critical for various applications, typically employing a two-step estimation process where intermediate parameters are first estimated based on different physical measurements, which are then used to determine positions. This includes methods leveraging doppler frequency shifts [1] for iterative determination. time difference of arrival (TDOA) measurements are a common basis for localization [3], [4], [7], often coupled with advanced estimation techniques or geometry optimization. For direction of arrival (DOA), approaches focus on multi-source localization [6] and advanced estimation for non-circular signals using array processing [8]. Additionally, tensor-based processing can be applied for passive localization of wideband emitters [5], and 5G networks are being explored for enabling cooperative vehicle localization [2]. While this decoupled two-step strategy simplifies computation, processing parameters independently can lead to suboptimal performance.

To address these limitations, Weiss introduced an alternative method known as direct position determination

(DPD) [9]. Unlike the two-step approach, DPD jointly processes the raw signals from all stations to estimate the source position directly, thereby avoiding information loss and achieving superior accuracy [10]. Despite its advantages, implementing DPD presents algorithmic challenges. The most widely used DPD estimator is based on the maximum likelihood (ML) criterion, but it involves a computationally intensive multi-dimensional search when multiple sources are present [11]. To mitigate computational costs, subspace-based methods derived from multiple signal classification (MUSIC) theory have been proposed, though they require prior knowledge of the source number [12]. Alternatively, the minimum variance distortionless response (MVDR) method is suitable for multi-source scenarios, but it often fails to converge to the Cramér-Rao lower bound (CRLB) as the mixture of signals may interfere with each other [13]. Recent advancements have sought to improve robustness and accuracy in these complex environments; for example, by exploiting the fourth-order cumulants of high-order QAM signals within a dual atomic norm minimization framework to suppress noise and handle multipath without knowing the source count [14], or by utilizing parallel factor analysis in the cross-correlation domain to expand virtual nodes for high-precision localization with reduced complexity [15].

This paper proposes a direct localization algorithm based on a multi-scale feature fusion network (MSFF) to address the problem of low localization accuracy under low SNR in complex environments. This neural network utilizes two innovations to enhance localization performance:

1. By learning source features in low signal-to-noise ratio (SNR) environments via multi-scale features extracted from the covariance matrix, the network achieves greater robustness compared to traditional algorithms and convolutional neural networks (CNN).

2. Unlike existing neural networks that often model such problems as multi-label classification tasks, the proposed multi-scale feature fusion neural network models direct

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localization as a regression task to achieve off-grid estimation, thereby attaining higher resolution.

II. System Model

Consider a direct localization scenario consisting of M spatially distributed base stations and N far-field narrowband sources. Let the position of the m -th station be denoted by $\mathbf{p}_m = [x_m, y_m]^T$ for $m = 1, \dots, M$. The state vector of the n -th source at a specific time instant is given by $\mathbf{u}_n = [x_n^{(s)}, y_n^{(s)}]^T$, where $(x_n^{(s)}, y_n^{(s)})$ represents the position.

Each base station is equipped with a uniform linear array (ULA) consisting of P sensors, and the inter-element spacing is assumed to be half-wavelength (i.e., $d = \lambda/2$). The direction of arrival (DOA) of the k -th source signal impinging on the m -th station is determined by the geometry between the source and the station. The sine of the arrival angle, denoted as $\sin(\theta_{m,n})$, is modeled as:

$$\sin(\theta_{m,n}) = \frac{y_n^{(s)} - y_m}{\|\mathbf{p}_n^{(s)} - \mathbf{p}_m\|_2}, \quad (1)$$

where $\|\cdot\|_2$ denotes the Euclidean norm.

At the m -th station, the array manifold vector corresponding to the n -th source is given by:

$$\mathbf{a}(\theta_{m,n}) = [1, e^{-j\pi \sin \theta_{m,n}}, \dots, e^{-j\pi(P-1) \sin \theta_{m,n}}]^T. \quad (2)$$

Assuming L snapshots are collected, the received signal matrix $\mathbf{X}_m \in \mathbb{C}^{P \times L}$ at the m -th station can be expressed as:

$$\mathbf{X}_m = \mathbf{A}_m \mathbf{S} + \mathbf{N}_m, \quad (3)$$

Here, $\mathbf{A}_m = [\mathbf{a}(\theta_{m,1}), \dots, \mathbf{a}(\theta_{m,N})] \in \mathbb{C}^{P \times N}$ denotes the steering matrix for the m -th station, while $\mathbf{S} \in \mathbb{C}^{N \times L}$ represents the source signal matrix, which is assumed to consist of Gaussian random processes. Additionally, $\mathbf{N}_m \in \mathbb{C}^{P \times L}$ is defined as the additive white Gaussian noise (AWGN) present at the m -th station.

The covariance matrix of the received signal at the m -th station is estimated by:

$$\hat{\mathbf{R}}_{xx,m} = \frac{1}{L} \mathbf{X}_m \mathbf{X}_m^H. \quad (4)$$

III. The Proposed MSFF Algorithm

This section models direct localization as a regression task and proposes a new framework for a deep learning network. Section III-A introduces the input/output forms and data preprocessing methods; Section III-B focuses on describing the network architecture and the implementation process of direct localization.

A. Data Preprocessing and Labeling

Direct localization is modeled as a regression task. Consequently, the output vector of the network, denoted as $\hat{\mathbf{y}}$, is designed to have a length of $2N_s$, where N_s is

the number of sources. The output vector consists of the estimated Cartesian coordinates for all sources:

$$\hat{\mathbf{y}} = [\hat{x}_1^{(s)}, \hat{y}_1^{(s)}, \hat{x}_2^{(s)}, \hat{y}_2^{(s)}, \dots, \hat{x}_{N_s}^{(s)}, \hat{y}_{N_s}^{(s)}]^T. \quad (5)$$

To effectively extract spatial features from the received signals, a specialized transformation is applied to the covariance matrices before feeding them into the neural network.

Assume the system comprises $M = K^2$ base stations, and each station is equipped with a linear array of P elements. Let $\mathbf{R}_i \in \mathbb{C}^{P \times P}$ denote the sample covariance matrix of the i -th station, where $i = 1, \dots, K^2$. The collection of these covariance matrices forms a tensor of dimension (K^2, P, P) .

Then, the real part, imaginary part, and phase of each covariance matrix are sequentially concatenated to form the input vector for the neural network, with dimensions of $(3, KP, KP)$. This decomposition results in a 3-channel input tensor $\mathcal{X} \in \mathbb{R}^{3 \times KP \times KP}$. The construction of the input tensor is defined as:

$$\mathcal{X} = \text{Stack}(\Re(\mathbf{R}_{global}), \Im(\mathbf{R}_{global}), \angle(\mathbf{R}_{global})), \quad (6)$$

where $\mathbf{R}_{global}$ represents the spatially stitched covariance matrix, which facilitates better feature extraction from the covariance matrices.

B. The Proposed Network Architecture

First, the specially processed covariance matrix is input into the multi-scale dilated convolution module and the multi-scale variable sampling aggregation module, respectively. Finally, the features extracted by the two networks are fused through the multi-scale feature fusion module. After passing through several fully connected layers, the final output directly represents the estimated results of the source coordinates. Fig. 1 illustrates the neural network structure.

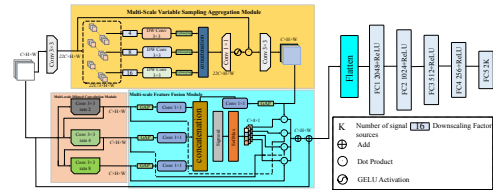


Fig. 1. The neural network structure.

a) Multi-Scale Variable Aggregation Module:

This module is designed to capture multi-scale contextual information from the covariance matrix elements while preserving fine-grained spatial details. Let the input feature map, which consists of covariance matrix elements upsampled via a 3×3 convolution, be denoted as $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$.

To extract features at different scales, the input $\mathbf{X}$ is first divided into three distinct groups along the channel dimension:

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3], \quad (7)$$

where each group is defined as $\mathbf{X}_i \in \mathbb{R}^{H \times W \times \frac{C}{3}}$ for $i \in \{1, 2, 3\}$.

Subsequently, each group is fed into a separate branch containing a max-pooling layer with a distinct window size. This design facilitates the aggregation of context from different receptive fields. Following the pooling operation, bilinear interpolation is applied to upsample the feature maps back to their original spatial resolution:

$$\mathbf{Y}_i = \text{UpSample}(\text{MaxPool}_{k_i}(\mathbf{X}_i)), \quad (8)$$

where k_i represents the pooling window size for the i -th group (e.g., 4×4 , 8×8 , and 16×16).

The resulting multi-scale features are then concatenated along the channel dimension to form a unified representation $\mathbf{Y}_{\text{concat}}$. To effectively integrate these multi-scale contexts with the original spatial information, an element-wise multiplication is performed between the concatenated features and the original input $\mathbf{X}$. This residual-like modulation aids in retaining spatial details that might otherwise be lost during pooling:

$$\mathbf{Y}_{\text{concat}} = \text{Concat}(\mathbf{Y}_1, \mathbf{Y}_2, \mathbf{Y}_3), \quad (9)$$

$$\mathbf{Y}_4 = \mathbf{Y}_{\text{concat}} \oplus \mathbf{X}, \quad (10)$$

where $\oplus$ denotes hadamard product.

Finally, a convolutional layer is utilized to fuse the channel information and restore the feature map dimensions to the initial state, yielding the final output $\mathbf{Z}_1$:

$$\mathbf{Z}_1 = \text{Conv}(\mathbf{Y}_4). \quad (11)$$

b) Multi-scale Dilated Convolution Module:

To effectively capture multi-scale contextual information, a cascaded dilated convolution architecture is utilized. Let the basic convolutional unit be denoted as $\mathcal{F}(\cdot; r)$, which consists solely of a 3×3 dilated convolution with a dilation rate r . This linear operation can be mathematically expressed as:

$$\mathcal{F}(\mathbf{X}; r) = \text{Conv}_{3 \times 3}^r(\mathbf{X}), \quad (12)$$

where $\mathbf{X}$ represents the input feature map.

Unlike standard parallel structures, this module adopts a progressive feature fusion strategy. The extraction process is conducted in a sequential manner, where the original input $\mathbf{X}$ is injected into each stage to preserve raw spatial details.

First, the input is processed by the first branch with dilation rate r_1 :

$$\mathbf{Z}_2 = \mathcal{F}(\mathbf{X}; r_1). \quad (13)$$

Subsequently, for the following stages, the input to the convolutional block is the element-wise summation of the previous stage's output and the original input $\mathbf{X}$. This connection ensures that both processed semantic features and original detailed features are fused before entering the next receptive field scale:

$$\mathbf{Z}_3 = \mathcal{F}(\mathbf{Z}_2 + \mathbf{X}; r_2), \quad (14)$$

$$\mathbf{Z}_4 = \mathcal{F}(\mathbf{Z}_3 + \mathbf{X}; r_3), \quad (15)$$

where r_1, r_2, r_3 correspond to the dilation rates of the three consecutive layers, and $\mathbf{Z}_2, \mathbf{Z}_3, \mathbf{Z}_4$ represent the intermediate feature maps extracted at each stage.

c) Multi-scale Feature Fusion Module:

This module employs a channel attention mechanism to adaptively fuse features from multiple scales. Let the input feature maps from different branches be denoted as $\mathbf{F}_i \in \{\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_3, \mathbf{Z}_4\}$, where each $\mathbf{F}_i \in \mathbb{R}^{H \times W \times C}$.

Global average pooling (GAP) is applied to each feature map $\mathbf{F}_i$, followed by 1×1 convolution to obtain a channel descriptor $\mathbf{g}_i$:

$$\mathbf{g}_i = \text{Conv}_{1 \times 1}(\text{GAP}(\mathbf{F}_i)), \quad (16)$$

where $\mathbf{g}_i \in \mathbb{R}^{1 \times 1 \times C}$.

The descriptors are concatenated along a new dimension to form a composite vector $\mathbf{G}$. Subsequently, a Sigmoid function is applied, followed by a Softmax function along the scale dimension to generate normalized weights α :

$$\mathbf{G} = [\mathbf{g}_{z1}, \mathbf{g}_{z2}, \mathbf{g}_{z3}, \mathbf{g}_{z4}], \quad (17)$$

$$\alpha = \text{Softmax}(\sigma(\mathbf{G})). \quad (18)$$

where σ denotes the Sigmoid activation function, and $\alpha = [\alpha_{z1}, \alpha_{z2}, \alpha_{z3}, \alpha_{z4}]$ represents the attention weights for the respective scales.

The original feature maps are weighted by their corresponding attention scores α_i via element-wise multiplication and then summed together. This aggregated feature $\mathbf{Z}_5$ is added to the original input $\mathcal{X}$ (residual connection) and passed through a final projection layer to obtain the output $\mathbf{Z}_6$:

$$\mathbf{Z}_5 = \alpha_{z1} \oplus \mathbf{Z}_1 + \alpha_{z2} \oplus \mathbf{Z}_2 + \alpha_{z3} \oplus \mathbf{Z}_3 + \alpha_{z4} \oplus \mathbf{Z}_4, \quad (19)$$

$$\mathbf{Z}_6 = \text{Conv}_{1 \times 1}(\mathbf{Z}_5 + \mathcal{X}). \quad (20)$$

d) Direct Position Determination Through Regression:

The module implements a multi-layer perceptron (MLP) regressor designed to predict offset values. Let the input feature map be denoted as $\mathbf{Z}_6$ and let N_s denote the number of sources.

The input tensor is flattened along the channel and spatial dimensions to form a feature vector $\mathbf{v}_0$:

$$\mathbf{v}_0 = \text{Flatten}(\mathbf{Z}_6). \quad (21)$$

The feature vector passes through a sequence of four hidden layers to extract high-level semantic information. For each layer $l \in \{1, 2, 3, 4\}$, the output $\mathbf{v}_l$ is computed as:

$$\mathbf{v}_l = \text{ReLU}(\mathbf{W}_l \mathbf{v}_{l-1}), \quad (22)$$

where $\mathbf{W}_l$ is the weight matrix, and $\text{ReLU}(\cdot)$ denotes the Rectified Linear Unit activation function. The dimensions of the hidden layers are progressively reduced as $d_1 = 2048, d_2 = 1024, d_3 = 512$, and $d_4 = 256$.

The final output vector $\mathbf{Z}_{\text{predict}}$, representing the predicted values:

$$\mathbf{Z}_{\text{predict}} = \mathbf{W}_5 \mathbf{v}_4. \quad (23)$$

where the output dimension is set to $2N_s$.

IV. Simulation And Experimental Results

This section presents the simulation setup and evaluates the performance of the proposed neural network (MSFF) based direct localization method. Comparisons are made against a CNN and the classical MUSIC algorithm under varying SNR levels and snapshot numbers.

A. Simulation Setup

A dataset is generated covering SNRs from -10 to 0 dB, where $N_s = 3$ sources are randomly positioned within $[250, 300] \times [250, 300]$ meters with $L = 500$ snapshots. The real, imaginary, and phase components of the sample covariance matrix (SCM) serve as input features. Partitioned into 90% training and 10% validation sets, the model is trained using PyTorch on an RTX 4090 GPU. The training runs for 100 epochs with a batch size of 256, utilizing the Adam optimizer with an initial learning rate of 1×10^{-4} .

The simulation environment models a multi-station scenario with $N_s = 3$ fixed sources at $(258.0, 254.0)$, $(281.0, 261.0)$, and $(264.0, 277.0)$ meters. The system comprises $M = K^2 = 4$ stations located at $(250, 300)$, $(300, 250)$, $(200, 200)$, and $(300, 300)$ meters. Each station consists of $P = 16$ uniformly spaced linear array elements with an inter-element spacing of $d = \lambda/2$ meters.

The localization performance is quantified using the root mean square error (RMSE) of the predicted source coordinates. For N_s true sources and their corresponding estimated locations over N_T trials, the RMSE is calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{N_T N_s} \sum_{t=1}^{N_T} \sum_{k=1}^{N_s} \|\mathbf{p}_{k,t}^{\text{est}} - \mathbf{p}_k^{\text{true}}\|_2^2}, \quad (24)$$

where $\mathbf{p}_k^{\text{true}} = [x_k^{\text{true}}, y_k^{\text{true}}]^T$ represents the true coordinate vector of the k -th source, and $\mathbf{p}_{k,t}^{\text{est}} = [x_{k,t}^{\text{est}}, y_{k,t}^{\text{est}}]^T$ denotes its estimated coordinate vector in trial t .

B. Performance Evaluation under Varying SNR

This experiment investigates the localization accuracy as a function of SNR. The number of snapshots L is fixed at 500. The SNR levels are varied from -12 dB to -2 dB in 1 dB increments. For each SNR level, $N_T = 500$ independent Monte Carlo trials are conducted. The RMSE for each model (MSFF, CNN, and MUSIC) is calculated at each SNR level.

As shown in Fig. 2, the RMSE generally decreases with increasing SNR for all methods. The proposed MSFF model consistently achieves lower RMSE values compared to the CNN and the MUSIC algorithm across the evaluated SNR range, demonstrating its superior robustness in noisy environments.

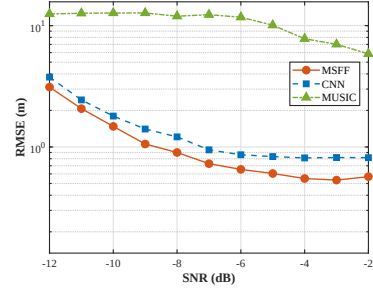


Fig. 2. RMSE versus SNR for the proposed model, CNN, and MUSIC algorithm ($L = 500$).

C. Performance Evaluation under Varying Snapshot Numbers

This experiment evaluates the impact of the number of snapshots L on localization accuracy. The SNR is fixed at -2 dB. The snapshot numbers are varied from 50 to 500 (specifically 50, 100, 200, 300, 400, 500). For each snapshot number, $N_T = 500$ independent Monte Carlo trials are conducted.

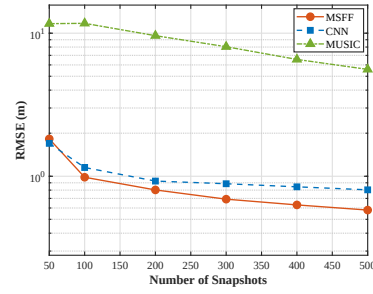


Fig. 3. RMSE versus Snapshot Number for the proposed model, CNN, and MUSIC algorithm (SNR = -2 dB).

Fig. 3 shows that increasing snapshots improves localization accuracy for all methods. While the proposed model slightly underperforms the CNN at $L = 50$, it surpasses the CNN at $L \geq 100$, consistently maintaining lower RMSE values. Notably, both deep learning approaches significantly outperform the MUSIC algorithm across all snapshot counts, demonstrating their superiority in this context.

V. Conclusions

This work presents a novel neural network-based direct localization algorithm designed to address challenges in noisy environments. By effectively leveraging multi-scale features from covariance matrix elements and integrating them through multi-scale spatial feature fusion, the algorithm directly predicts source locations. Simulation results consistently demonstrate its superior localization accuracy and robustness compared to MUSIC and CNN.

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